

Empirical Proof Record: VALIDATED

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Proof ID: aa0dc369d169ac57c391a606be8809d8...

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1. Claim

Under sustained load, the substrate stays cognitively alive on real input: (for every cycle that produced a concept set, $\Phi \geq 0.05$). Empty-concept cycles are excluded — with nothing to integrate, $\Phi=0$ is correct, not a defect; their rate is reported separately. The floor is the lowest- Φ real-input cycle across the run.

- **Operationalisation:** Stream 3 passes over 8 substantive stimuli through Kimera’s entity target (Takwin); read the per-cycle “phi” and “concepts”. Partition cycles into real-input (concepts non-empty) and empty-input. $\text{phi_floor} = \min \Phi$ over real-input cycles; the LTL invariant holds iff that $\text{floor} \geq \varphi_floor$. empty_input_rate and the over-all-cycles phi_floor_strict are reported alongside.
- **Threshold:** $\text{phi_floor} \geq 0.05$ phi
- **H0:** H0: $\min \Phi < 0.05$ on real-input cycles — Φ collapses below the floor despite the input producing concepts
- **H1:** H1: $\min \Phi \geq 0.05$ on real-input cycles — the substrate stays cognitively alive whenever it has something to integrate

2. Verdict

VALIDATED — observed 0.6603084741552229

Φ floor 0.6603 over 24 real-input cycles (3 passes $\times$ 8 stimuli); mean Φ 0.6892, band [0.6603, 0.7023], stdev 0.0141; non-collapse rate 100.0%; 0 empty-input cycles excluded (0.0%; strict floor over all cycles 0.6603); 0 cycles produced no Φ ; worst real cycle: stimulus 1 @ cycle 1 = 0.6603; cross-check skipped: non-collapse Wilson 95% CI [0.862, 1.000]

3. Evidence

Pillar	Statistic	Value	95% CI	Library
0.flow.phi_stability	phi_floor	0.6603084741552229	—	ophamin 0.115.2

4. Pre-registration

- Registered at: 2026-05-23T23:12:36.297925+00:00
- Config hash: fc0d6fa2b433540d0fc987795a9290da...
- Data hash: 4e6090352a40ea27df1cf08fb4d02bd5...

5. Signature

5c19e91d8603fb985c4ac93ae11e5a835c47bea97430c0e790d098c59cf83791